COMP 3900 Computer Science Project 2023 T1

Proposal

Project title: Movie Finder System

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1. Background

During the outbreak of the coronavirus, some countries have adopted closed management measures (Esposito & Principi, 2020). With many people forced to work and live from home, movies have become a popular way to kill the time (Moon, 2020). At this time, a complete movie search and movie evaluation exchange platform has become what people urgently want. In the process of searching for the movies, disorganized movie information pages and irrelevant movie recommendations make it difficult for people to find the movies they are truly interested in. Communications about a particular movie in various forums are not aggregated, which can lead to confusion for people who want to see a review of this movie. Therefore, the purpose of our website is to provide a one-stop solution for all people's need for movie information, including user login system, movie basic information search system, movie evaluation and review system, movie recommendation system, director basic information search system, cast basic information search system, user communication system, Blacklist, Wishlist, and watched list system, management system for product owners, website security system.

Compared with other similar websites on the market, our website has many advantages. DouBan is one of the most popular movie finder platforms in China, which has the same problem area as the system we want to build. It is a movie, TV, music, and book search platform integrated with social functions. It can display various information about the search object, including introduction, author, screenshots, short reviews, and long reviews. Users can freely discuss and rate specific objects on this platform, which has caused the main point of controversy on this platform - the orientation of public opinion. It is said that the bulk of malicious reviews from regular users on DouBan can harm the film industry (Han et al., 2012). Our system provides a solution to this problem, with the ability for the administrator to tag users and the ability for the administrator to edit and delete multiple malicious comments.

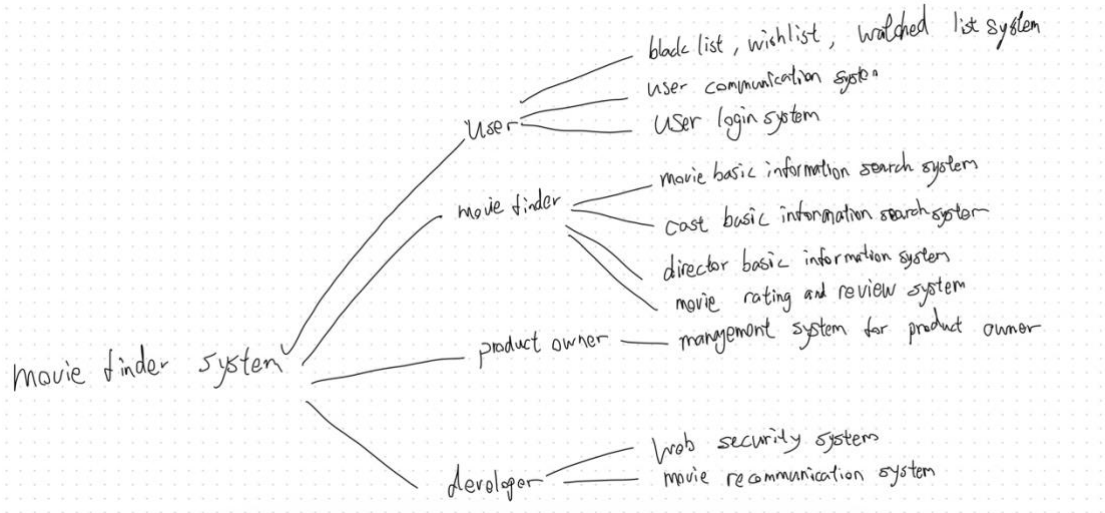
DianYingGou is another movie finder site used by many people in China. It can have a wide range of search functions, but it does not have a user login system. When a user wants to share a movie with a friend, it requires a third-party software to send a screenshot of the movie details page, which is cumbersome for the user. Our system has a user communication system that allows users to communicate with each other more easily. The communication system also helps users to have a more convenient way to meet people who share the same cinematic tastes.

Overall, our website can stand out from other similar websites on the market, offering a comprehensive movie search and evaluation exchange platform with advanced features that can enhance user experience and facilitate communication

among users.

2. User stories and Sprint

- **Structure**



Our entire system is made up of 10 subsystems. Each of their systems exists to satisfy one or more user stories. They include blacklist, Wishlist, watched list system, user communication system, user login system, movie basic information search system, cast basic information search system, director basic information system, movie rating and review system, management system, management system for product owner, web security system, movie recommunication system.

Each system has a specific revenue group. We have separated user and movie finder because some functions are available only if you are logged in, while others are available without logging in. product owner exists to ensure that the community is stable and does not have any serious consequences. developer exists to ensure that the site is stable, can withstand a large number of queries, and is protected against attacks.

- **User stories:**

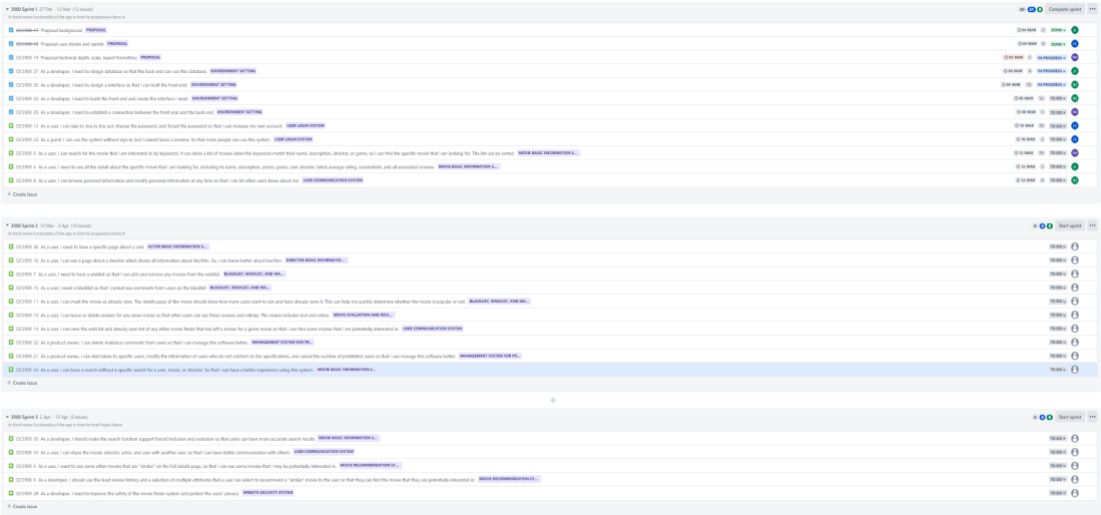
Role	User Story	Acceptance Criteria
User	I can sign in, log in, log out, change the password, and forget the password so that I can manage my own account. [user login system]	The system allows users to create a new account for themselves using their E-mail. Users can choose to log in with E-mail and password. User can change their passwords after they logged in. The system can help user to reset the password through email authentication when they forget the password.
User	I can browse personal	User information includes email,

	information and modify personal information at any time so that I can let other users know about me. [user communication system]	first name, last name, age, sex, occupation, area, description, tag, permission level. For new accounts that have just been created, some user information is empty. Users can update their personal information excludes tag, permission level in their account detail page.
User	I can use the system without signing in, but I cannot leave a review. So that more people can use this system. [user login system]	The system allows user to search the movie, see the review, see the detail of the movie, see the detail of the director, see the detail of the cast and see other users' account detail.
User	I need to have a Wishlist so that I can add and remove any movies from the Wishlist. [blacklist, Wishlist, watched list system]	User must be able to add and remove any movie to the Wishlist. User can be able to view other users' Wishlist in their information page.
User	I need a blacklist so that I cannot see comments from users on the Blacklist. [blacklist, Wishlist, watched list system]	Users must be able to add and remove users from the Blacklist.
User	I can mark the movie as already seen. The details page of the movie should show how many users want to see and have already seen it. This can help me quickly determine whether the movie is popular or not. [blacklist, Wishlist, watched list system]	User must be able to add and remove any movie they come across in the system to a watched list. The details page of the movie should show how many users want to see and have already seen it.
User	I can view the wish list and already seen list of any other movie finder that has left a review for a given movie so that I can find some movies	User can be able to view other users' Wishlist and already seen list in their information page.

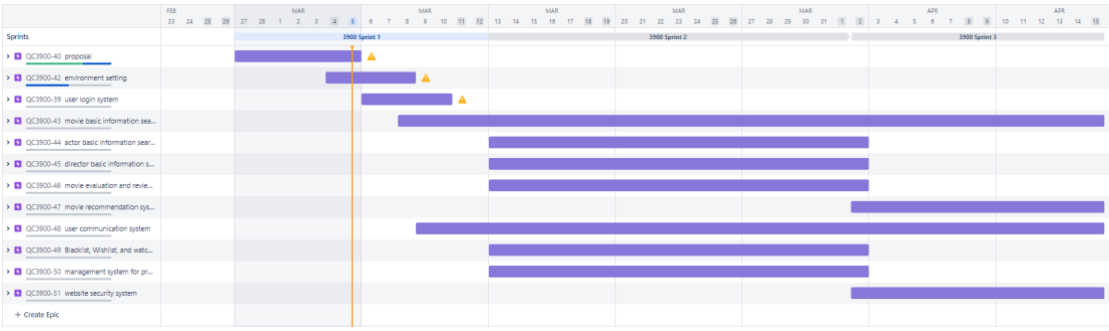
	that I am potentially interested in. [user communication system]	
User	I can share the movie, director, and user with another user, so that I can have better communication with others. [user communication system]	Each pair of friends can chat with each other and share movies, directors, casts, and users.
Movie finder	I need to see all the details about the specific movie that I am looking for, including its name, description, genre, cast, director, latest average rating, screenshots, and all associated reviews. [movie basic information search system]	Movie information includes movie name, release date, director, language, description, tag, list of casts, visit volume, rating, review, screen shot. List of casts is sorted by their participation.
Movie finder	I can search for the movie that I am interested in by keywords. It can show a list of movies when the keywords match their name, description, director, or genre, so I can find the specific movie that I am looking for. The list can be sorted. [movie basic information search system]	Once the keyword exactly matches part of the movie name, the movie should add into the list. The result list can be sorted by release date, alphabetical order, rating, number of reviews, number of views.
Movie finder	I need to have a specific page about a cast. [cast basic information search system]	Information of cast includes name, birthday, sex, nation, views, number of performances, pictures.
Movie finder	I can see a page about a director which shows all the information about her/him. So, I can know better about her/him. [director basic information search system]	Information of director includes name, description, birthday, nation, sex, visit volume, number of performances.

Movie finder	I can leave or delete reviews for any given movie so that other users can see these reviews and ratings. The review includes text and rating. [movie rating and review system]	User must also be able to leave a review for any given movie they view, which includes text and a rating from 0 to 5. The review displayed on the detail page of the movie should include content, rating, release date, number of agree, number of disagree, author.
Movie finder	I can have a search without specific this search for a user, movie, or director. So that I can have a better experience using this system. [movie basic information search system]	User can choose whether to search for a movie, cast, director or user after the search results are available. On the movie search results page, each movie is displayed with the movie name, description, rating and number of reviews. On the cast search results page there are cast names, descriptions, views, and number of performances. On the director search results page there is the name of the director, description, views, number of performances. On the user search results page there is the name of the user, views, and the number of friends. The movie result list can be sorted by release date, alphabetical order, rating, number of reviews, number of visits. The cast result list can be sorted by alphabetical order, views, number of performances. The director result list can be sorted by alphabetical order, views, numbers of performances. The user result list can be sorted by alphabetical order, views, numbers of friends.
Movie finder	I want to see some other movies that are "similar" on	In the page of movie detail, there should be a part for the “similar”

	the full details page, so that I can see some movies that I may be potentially interested in. [movie recommendation system]	movie. The movies in this section should be relevant to the user's wish list and the movie they are currently viewing.
Product owner	I can delete malicious comments from users so that I can manage this software better. [management system for product owner]	Administrators have the right to delete comments in order to prevent malicious comments from leading public opinion. For the normal development of the system and the film industry, administrators have higher rights than ordinary users.
Product owner	I can Add labels to specific users, modify the information of users who do not conform to the specifications, and cancel the number of prohibited users so that I can manage this software better. [management system for product owner]	Some users may post comments pretending to be influencers, so it is necessary that only the administrator can add the user's authoritative tag.
Developer	I should make the search function support forced inclusion and exclusion so that users can have more accurate search results. [movie basic information search system]	If the user's input in the search box contains spaces, the default search result must contain all the keywords entered. If the user's input in the search box contains a "-", the default search result must not contain the word after the "-". Search system is case-insensitive.
Developer	I need to improve the safety of the movie finder system and protect the users' privacy. [website security system]	The website should be able to withstand a sudden influx of requests for access.

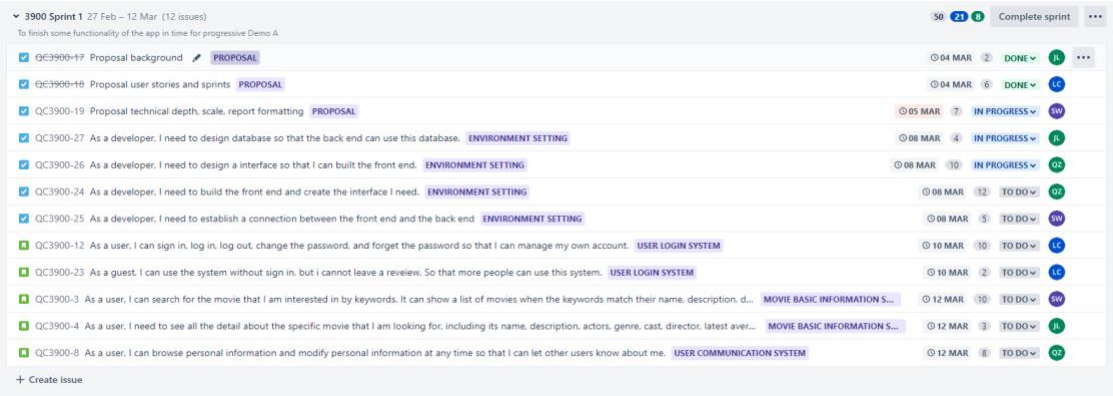


- **Sprints**



All product backlogs are divided into three sprints. As we can see on the graph, priority in order from highest to lowest.

The first sprint (27th February 2023 – 12th March 2023):

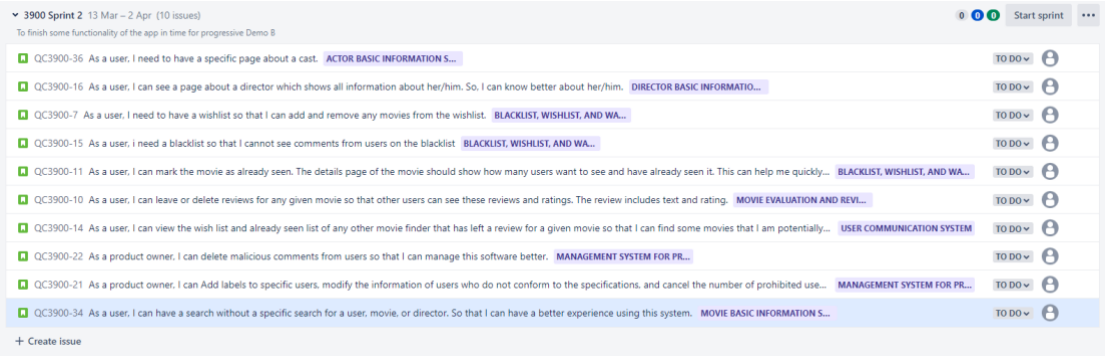


This sprint focuses on user sign in, login settings, basic movie search, and basic movie results. The user login system should be the first to be completed, as many functions need to be implemented after the user login. The function of searching movie, showing searching result and login settings can help us to structure the basic

user communication system and the movie basic information search system. They can make the system achieve the fundamental function of movie finder. At that point the complete database will be created which is a preparation for the next steps.

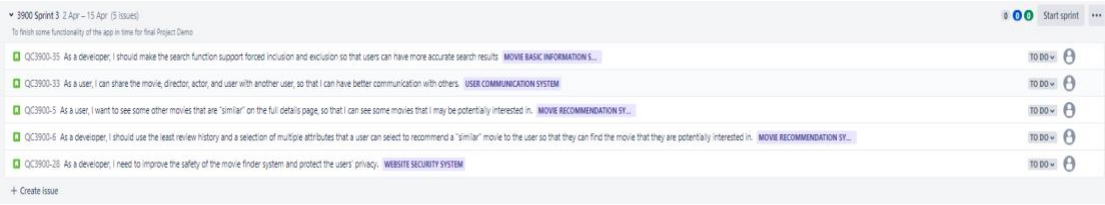
Deadline, Estimated Completion Time for each task/user story is on the graph.

The second sprint (13th March 2023 – 2nd April 2023):



For Demo B.

The third sprint (2nd April 2023 – 15th April 2023):



For final Demo.

- **Norvel function**

Movie basic information search system with forced inclusion and exclusion function

I think one of the most powerful features of Google search is about its settings search. Forced inclusion and forced exclusion is one of the most common search techniques I use. For example, if I want to search for World War II without including the USA, I just type in World War II – USA. I think all searches should include this feature, but I have found that most of the movie finding systems on the market do not include this feature. This can make it difficult to find the content I'm looking for.

The integration of movie basic information search system, cast basic information search system, director basic information search system, and user search information.

In many existing work or systems in the same problem domain, they always let the user choose whether this is a movie search or a director search or a user search or a cast search before searching. This is an extra step before the search which is bothersome and unwanted by the user. Eliminating this extra step is an important step in improving the user experience, which I think is important. On the search results page of our system, we can directly see all relevant objects, whether it is a movie, a cast, a director, or a user.

User communication system

In many existing work or systems in the same problem domain, they only provide all information about the film and nothing else. For example, DianYingGou only lets the user know that the film exists. If you want to share this information with your friends, you can only take a screenshot and send it to them via third party software. I think an easy and fast sharing and communication system is the key to attracting users. Making friends with people who share the same tastes and being able to share in real time on our system is important to our users.

3. Technical depth, scale, report formatting

- **System Architecture**

Interface Layer (CSS JavaScript HTML)
Interface Layer also known as User Interface Layer which is closest to the user. It is used to display data and receive data input by users, providing users with an interactive operation interface. In order to realize the Interface Layer, we will use HTML, CSS, and JavaScript, the three tools to create the responsive application. The interface will be built with a react framework. The react framework is very fast. It does not directly operate on the DOM, but introduces a concept called virtual DOM, which is inserted between the JavaScript logic and the actual DOM, with good performance.

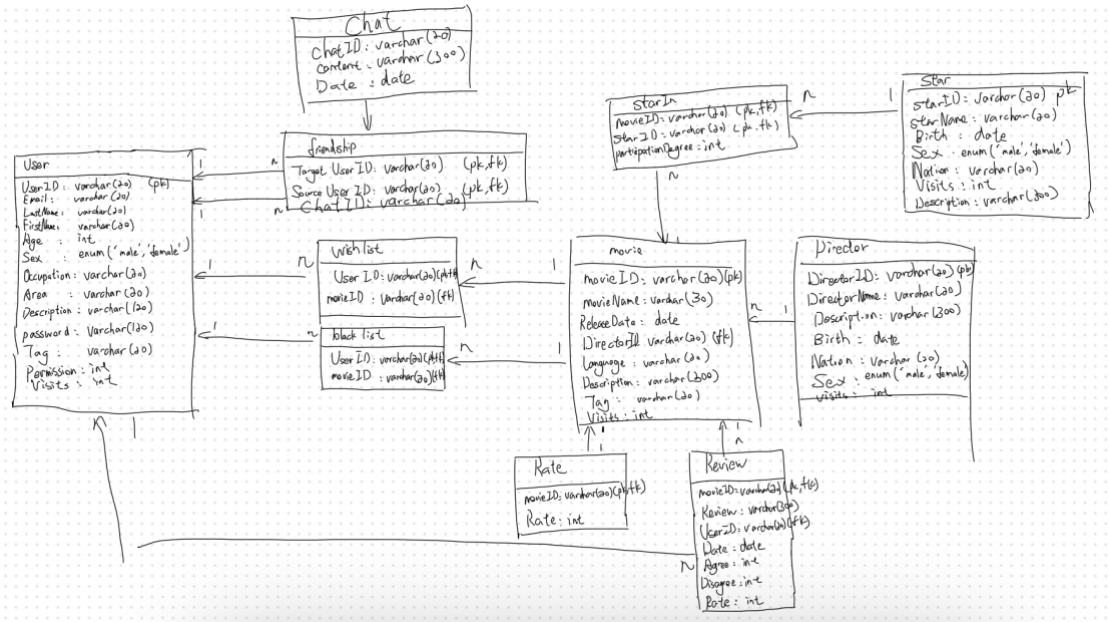
Business Layer (Python)
To finish the backend work, we will use the Python language, as Python is an efficient and convenient language. Python will allow us to use the Django framework to connect the frontend to the backend. Django is a lightweight open-source web application framework. Django has the following advantages. It contains many functional modules, which means that most of the things that developers want are included. Therefore, there is less need for third-party plugins.

API Layer (Python, MySQL-connector-python)
The MySQL connector/Python module is an official driver supported by Oracle for connecting to MySQL through Python. The connector is entirely Python, while MySQL client is written in C. It's also self-contained, meaning it doesn't require

the MySQL client library or any Python modules other than the standard library. Relatively speaking, this method is relatively simple and fast for the connection between the backend and the database.

Database Layer (MySQL)

For the database, we will choose the MySQL database to store our data. This database has a wide range of usage scenarios, and the language is low-complexity and easy to learn. MySQL is a relational database management system, which means that complex data can be stored in multiple tables and related to each other. It supports the manipulation of a wide variety of data types, including integers, strings. So, we can use the MySQL database to store users' personal information, movie information, director information, cast information, chat record, and reviews. Compared to using local JSON files to store data, MySQL can use multiple threads to make queries faster and the structure more flexible to modify. Based on our software requirements, we chose to use it.

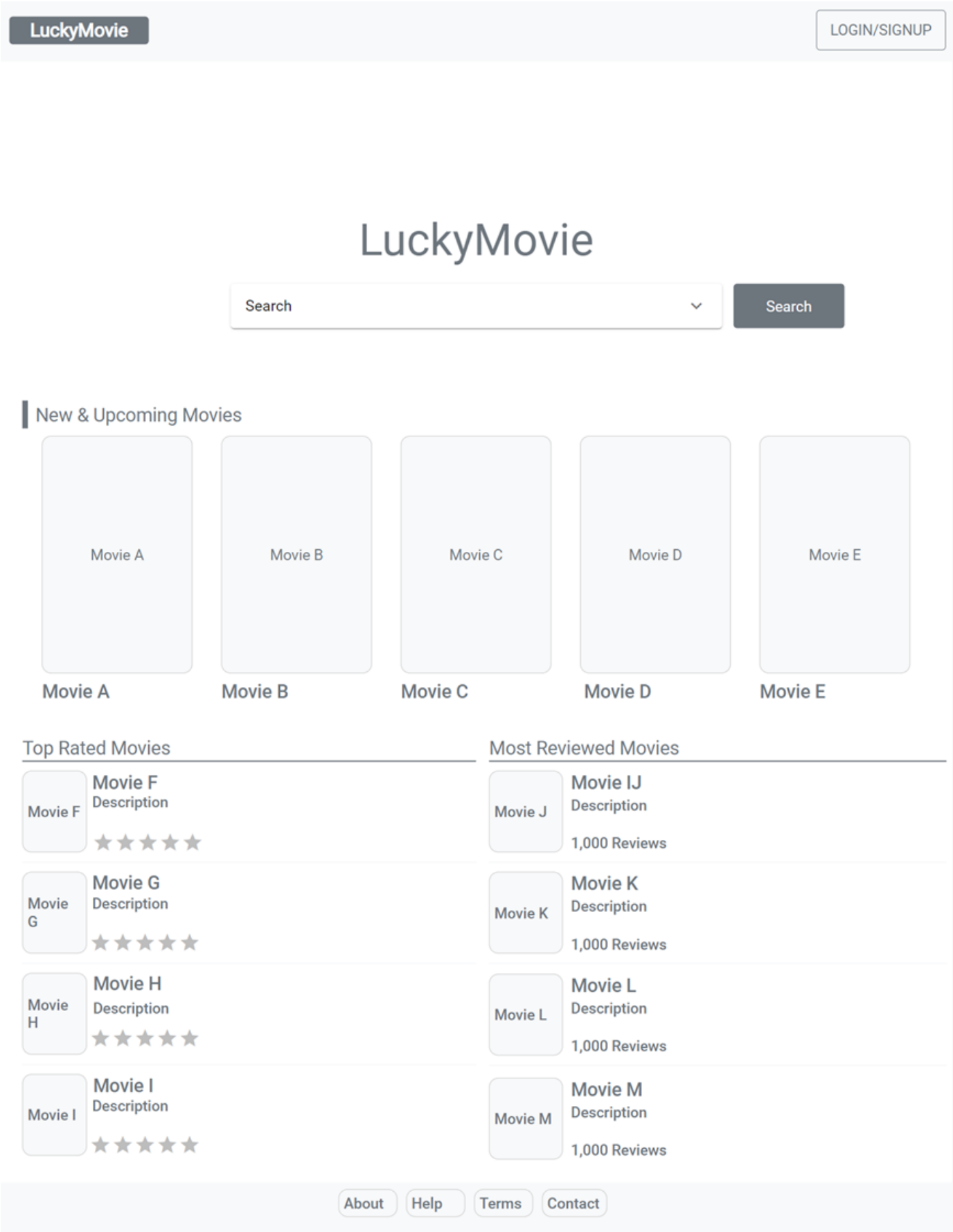


Infrastructure Layer (PyCharm)

PyCharm is an IDE dedicated to the python language, which has great compatibility with the backend. It has a set of tools that can help users improve their efficiency when developing in Python language, such as debugging, syntax highlighting, project management, code jumping, smart prompts, auto-completion, unit testing, and version control. In addition, the IDE provides advanced features for supporting professional web development under the Django framework.

- **Interface, flow diagrams and flow story**

Next, we will introduce the use process of the entire website. Whether you are a tourist or a person who wants to become a user of this website, you can enter this website to obtain movie information.


LuckyMovie
LOGIN/SIGNUP
LuckyMovie
Search
Search
New & Upcoming Movies
Movie A
Movie B
Movie C
Movie D
Movie E
Movie A
Movie B
Movie C
Movie D
Movie E
Top Rated Movies
Movie F
Movie F
Description
Movie G
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Movie L
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Movie M
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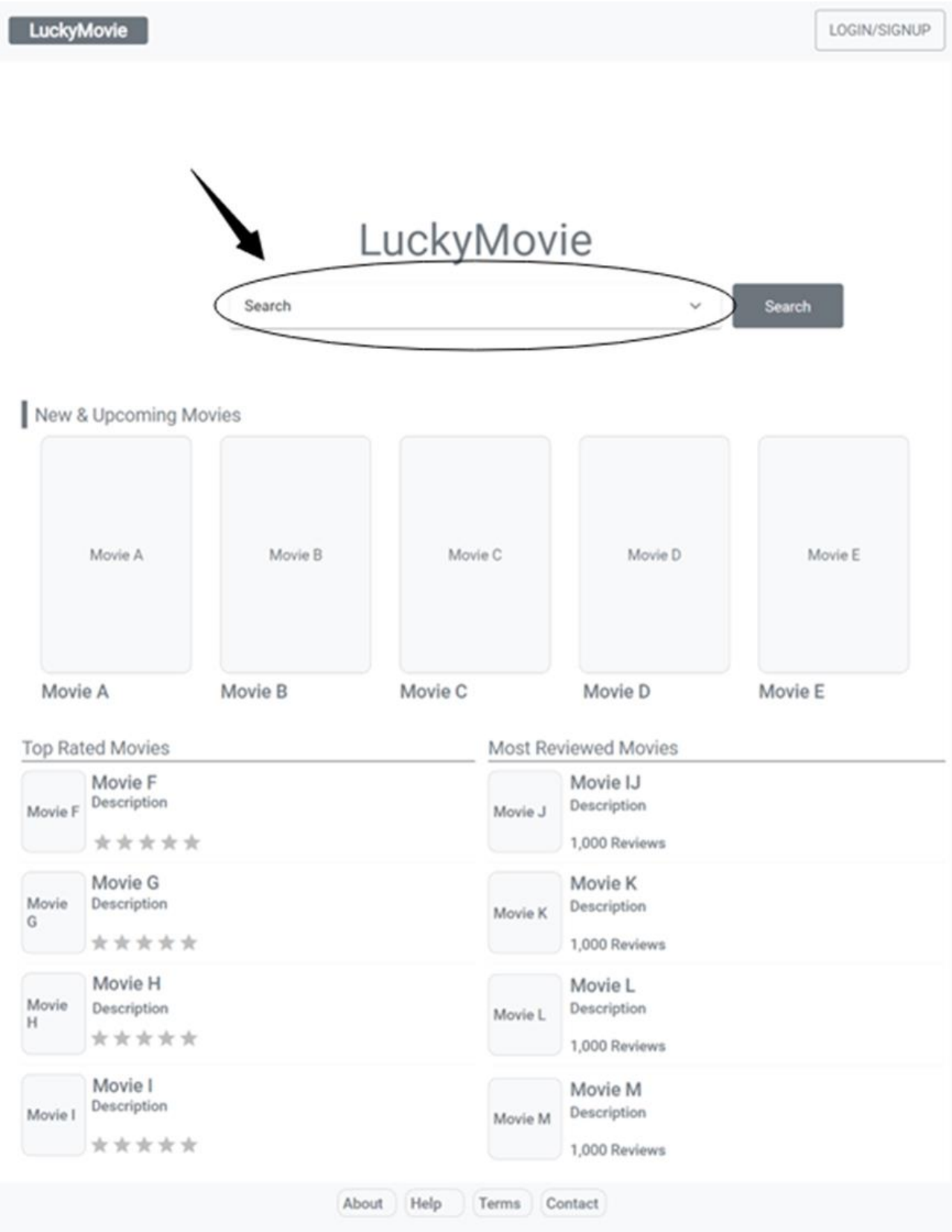
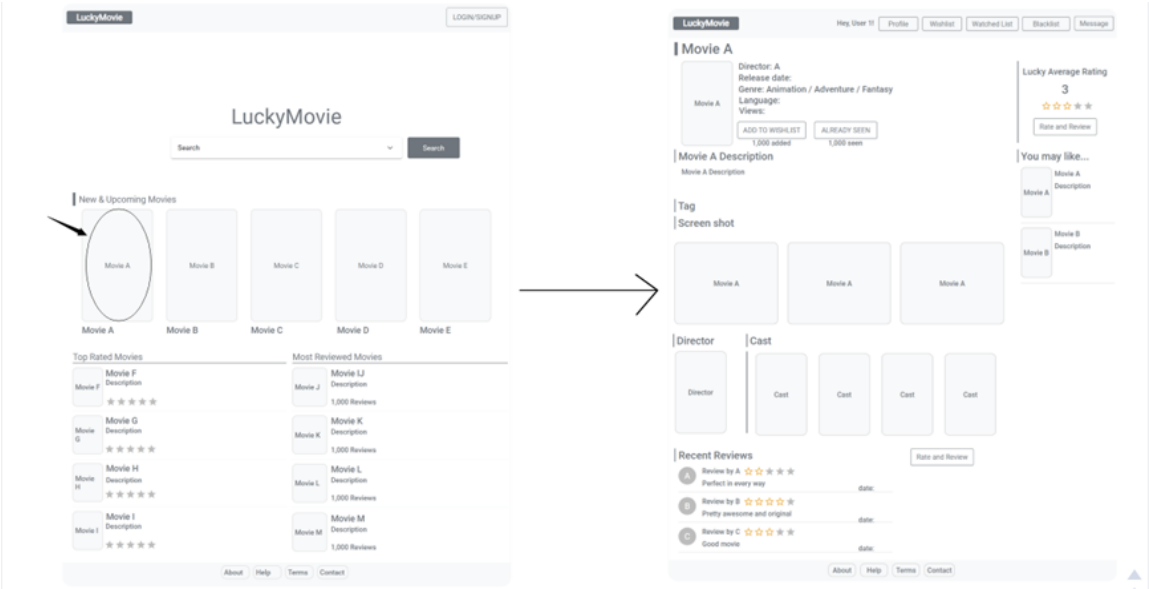
About
Help
Terms
Contact

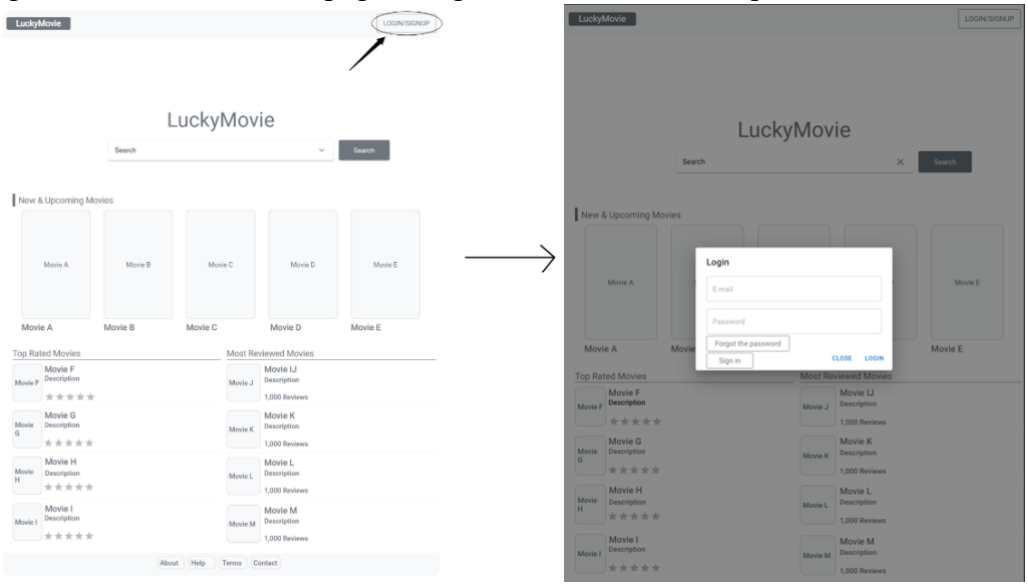
As a tourist, you can directly enter the name of the movie you want to know in the search bar.

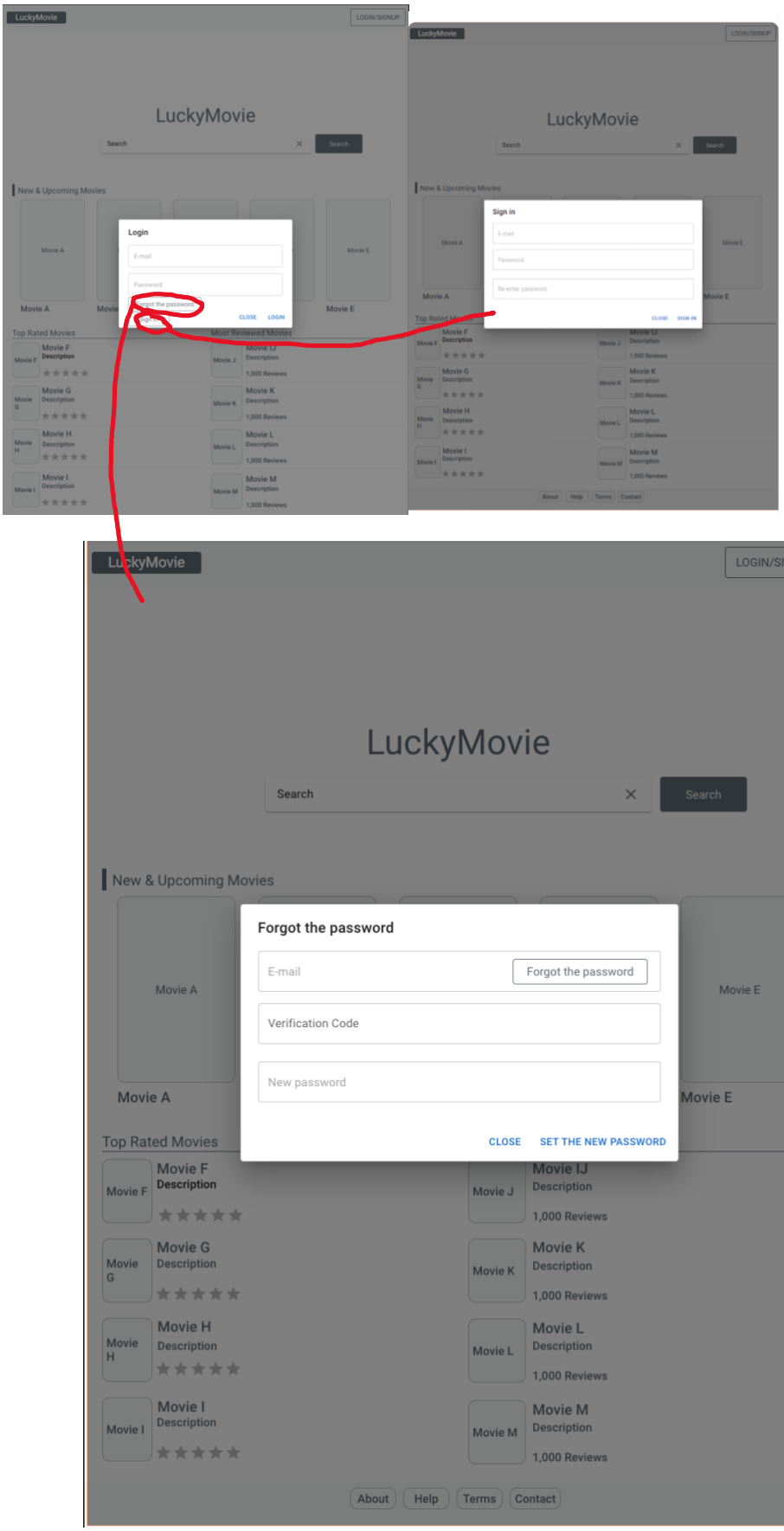


The initial page will also display popular movies at the moment. If you want to know about these movies, you can directly click on the poster of the current movie to know the movie information.



If you want to become a user, you need to click the login/signup button in the upper right corner of the current page to register. And set the new password.




LuckyMovie
LOGIN/SIGNUP
LuckyMovie
Search
Search
New & Upcoming Movies
Movie A
Movie E
Login
E-mail
Password
Forgot the password
CLOSE
LOGIN
Sign in
E-mail
Password
Re-enter password
CLOSE
SIGN IN
Top Rated Movies
Most Reviewed Movies
Movie F
Description
Movie G
Description
Movie H
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Movie I
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Movie J
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1,000 Reviews
Movie K
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Movie M
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Terms
Contact
LuckyMovie
LOGIN/SIGNUP
LuckyMovie
Search
Search
New & Upcoming Movies
Movie A
Movie E
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E-mail
Forgot the password
Verification Code
New password
CLOSE
SET THE NEW PASSWORD
Top Rated Movies
Movie F
Description
Movie G
Description
Movie H
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Movie I
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1,000 Reviews
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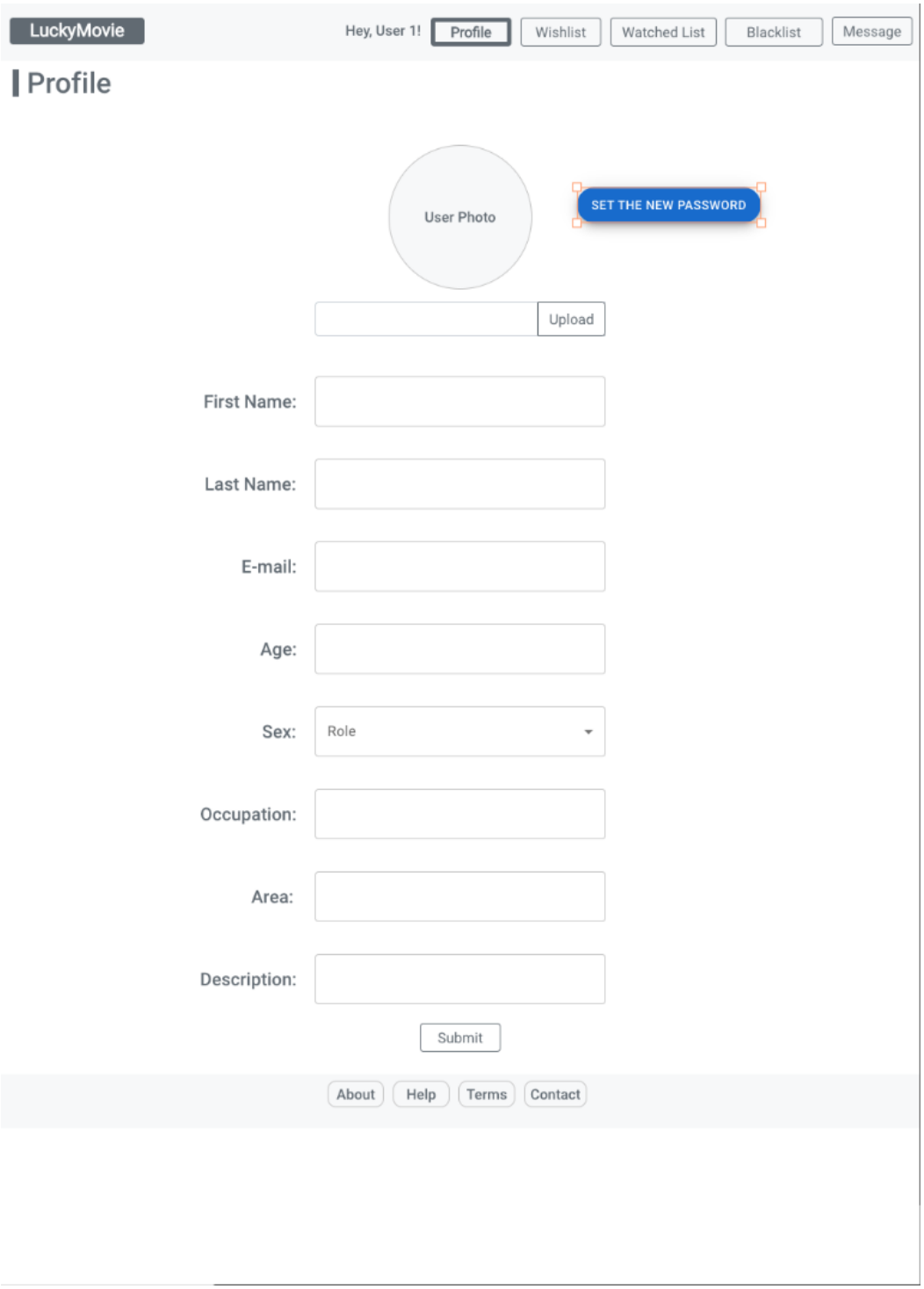
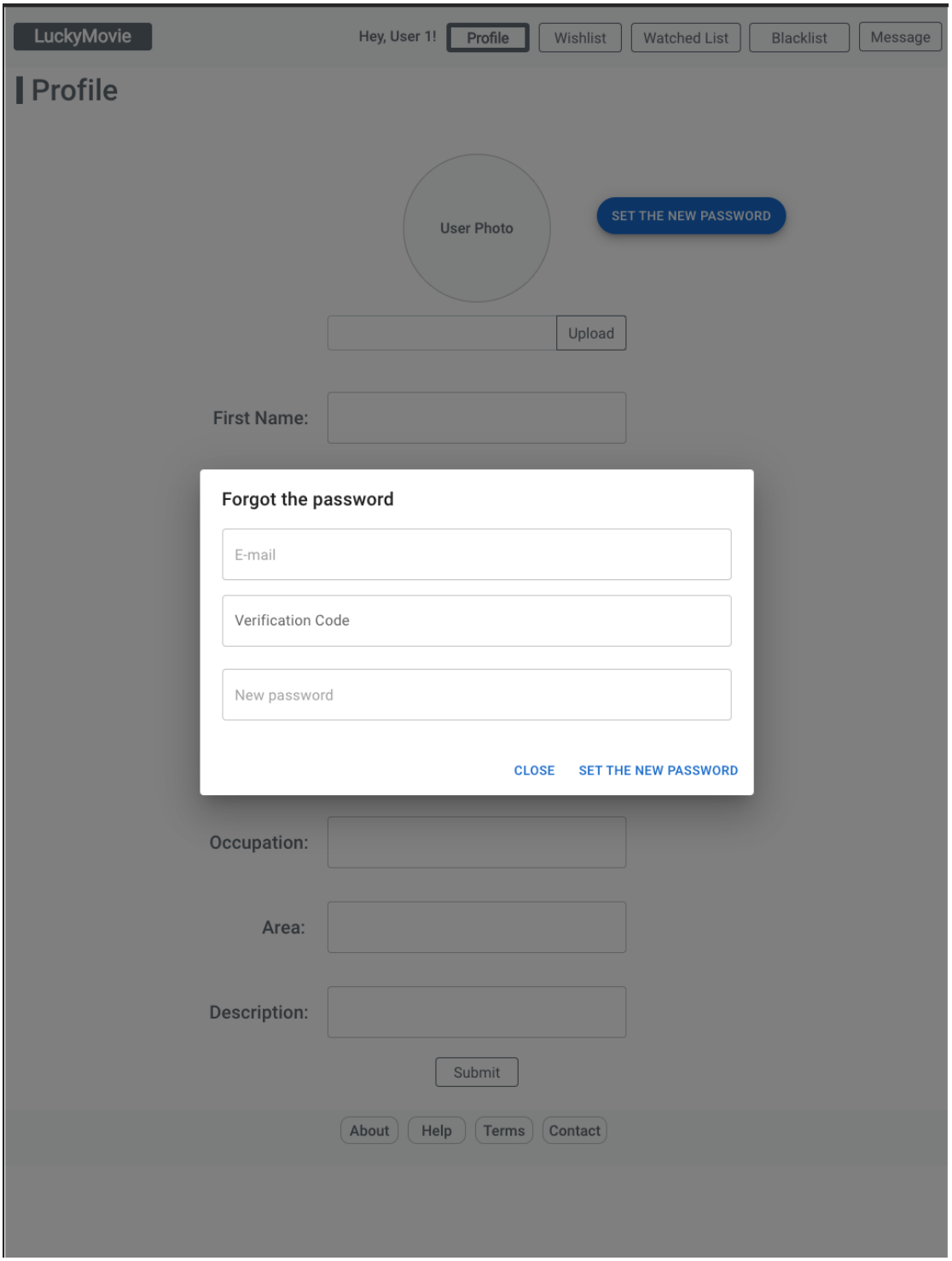
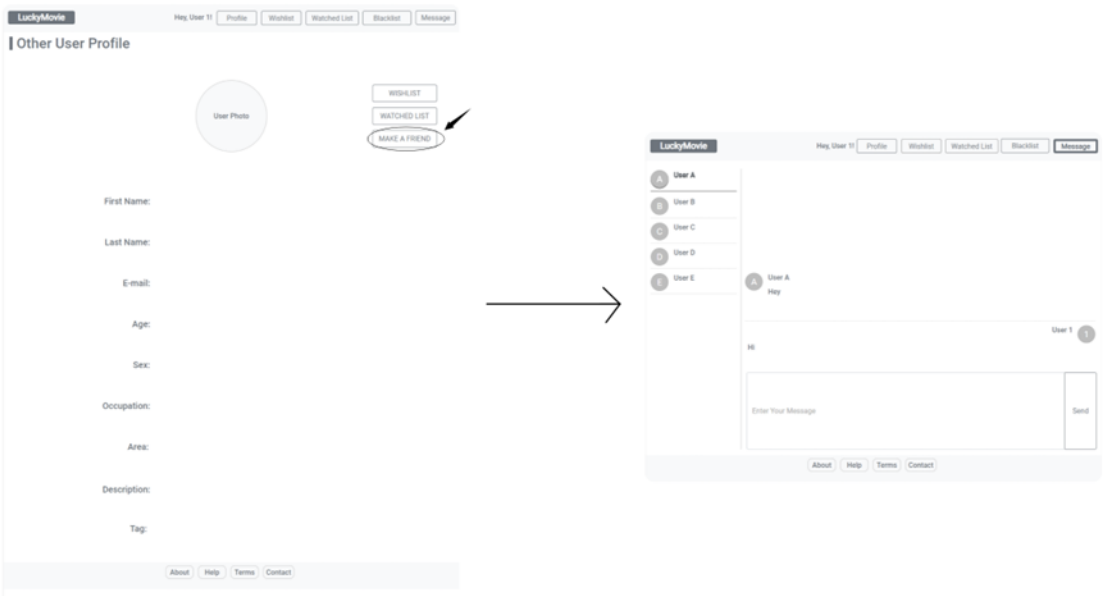
About
Help
Terms
Contact

After you register, you can log in every time you visit this website. As registered users, you have more functions than tourists. You can refine and modify your personal information,

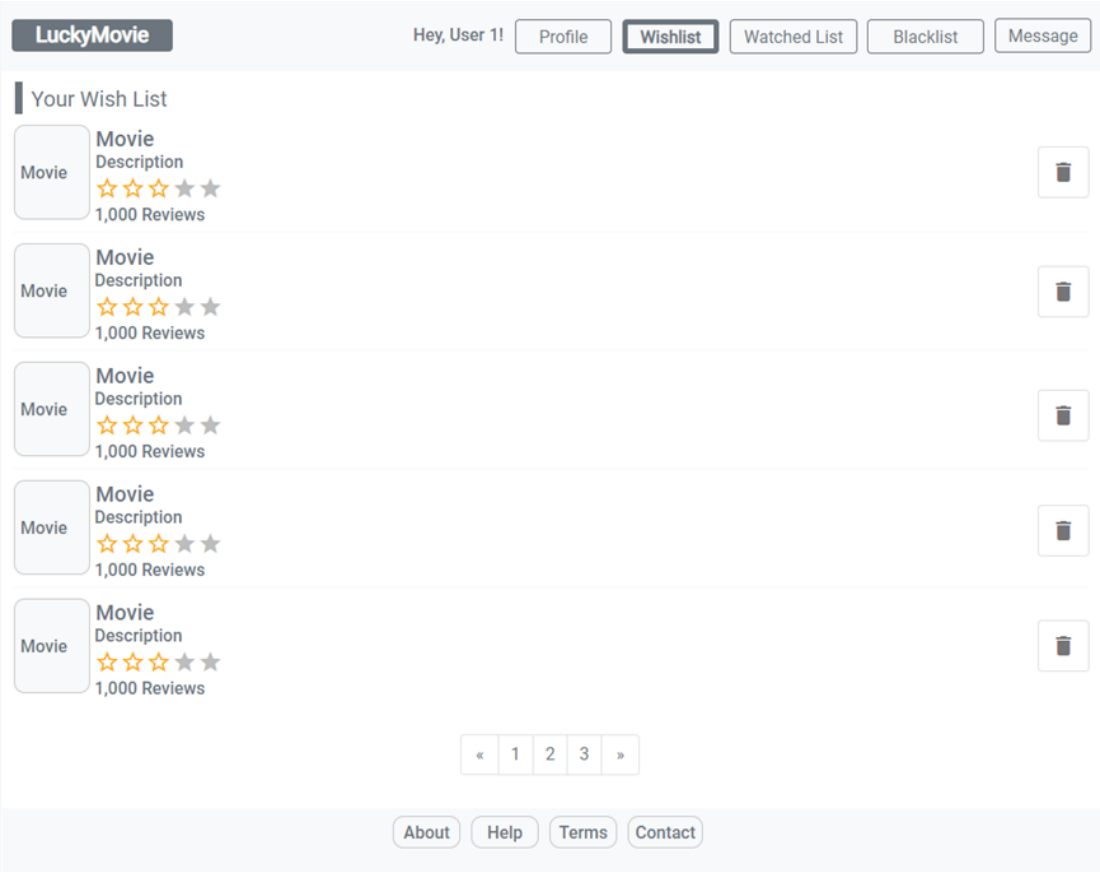




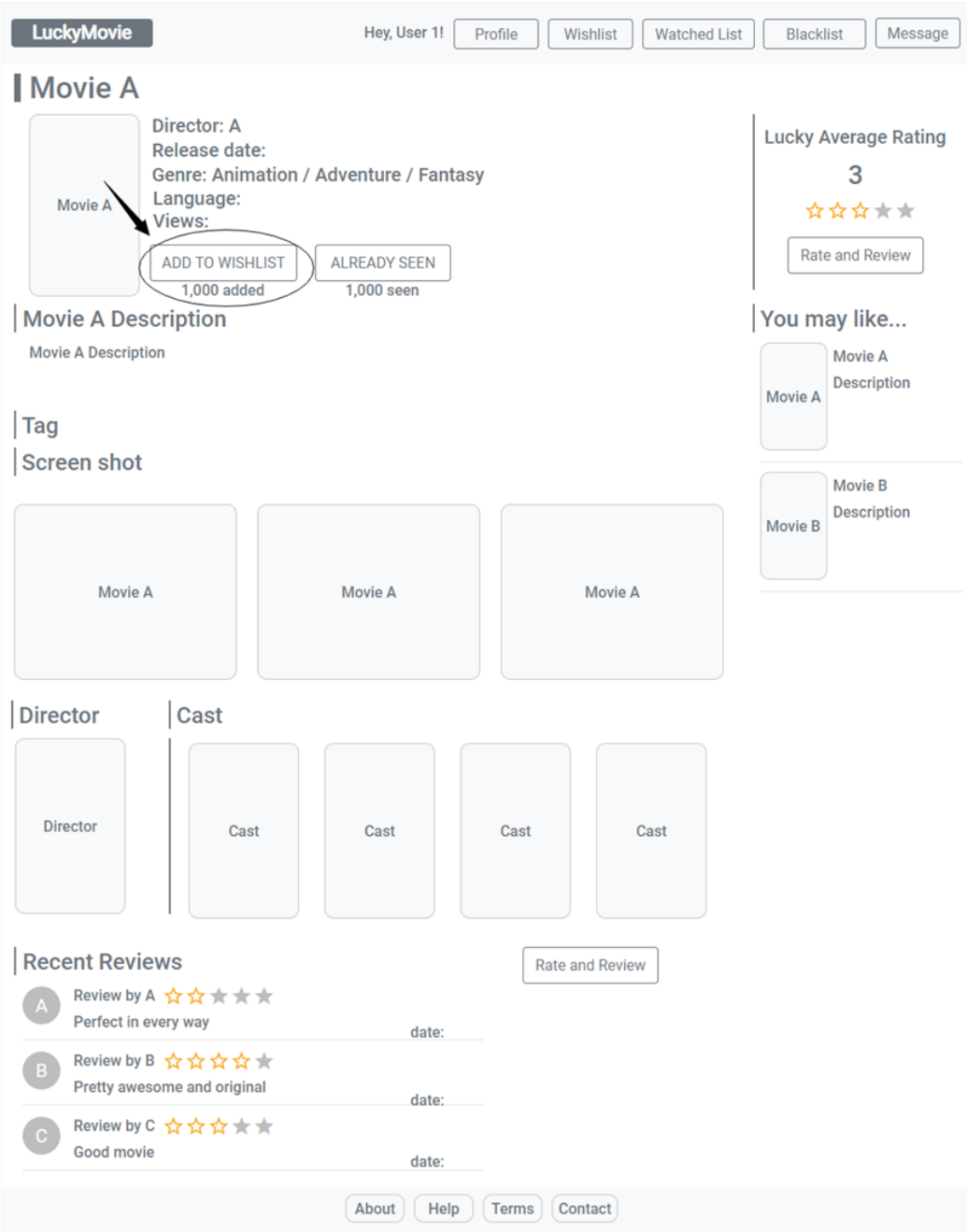
Our website has a novel function compare with other websites. If you find that a user's review is very interesting with you, you can enter the user's home page. There will be a button called make friend on the home page, you can click this button. Then you two users can discuss movies you both like.



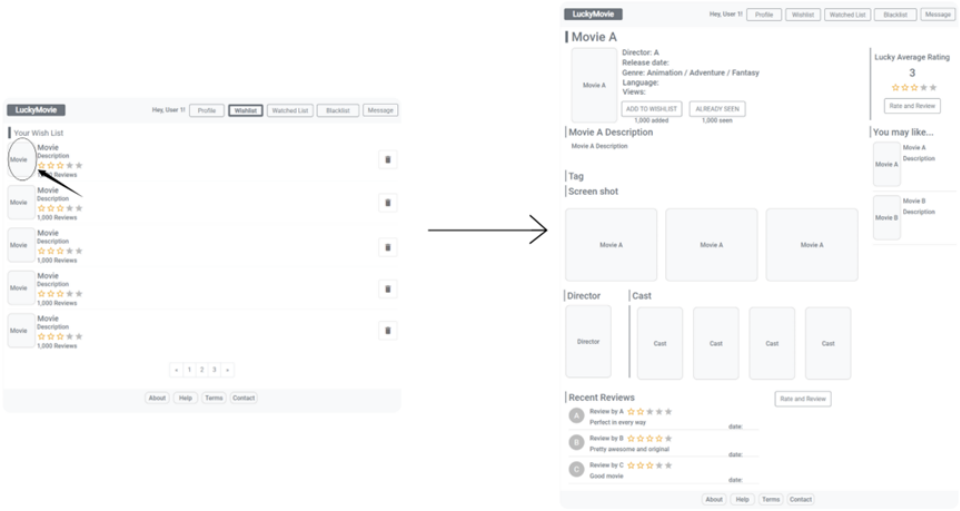
As a user, of course, you want to be able to access your favorite movies more quickly, so we have added a wish list for you.



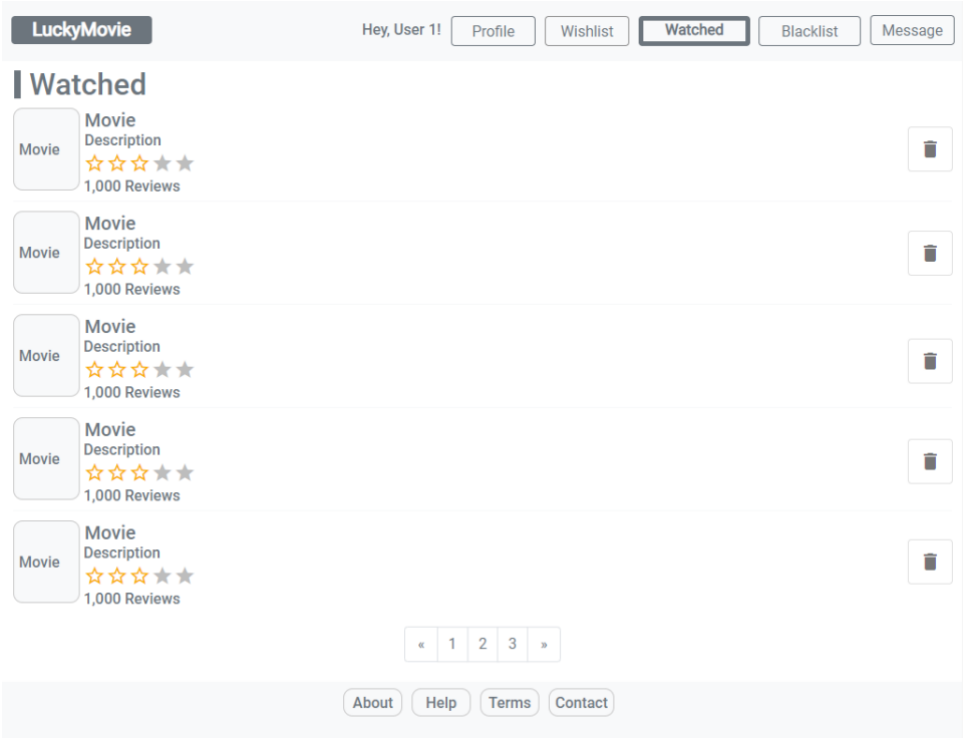
Whenever you visit a movie, if you find that you like this movie, you can add this movie to the wish list.



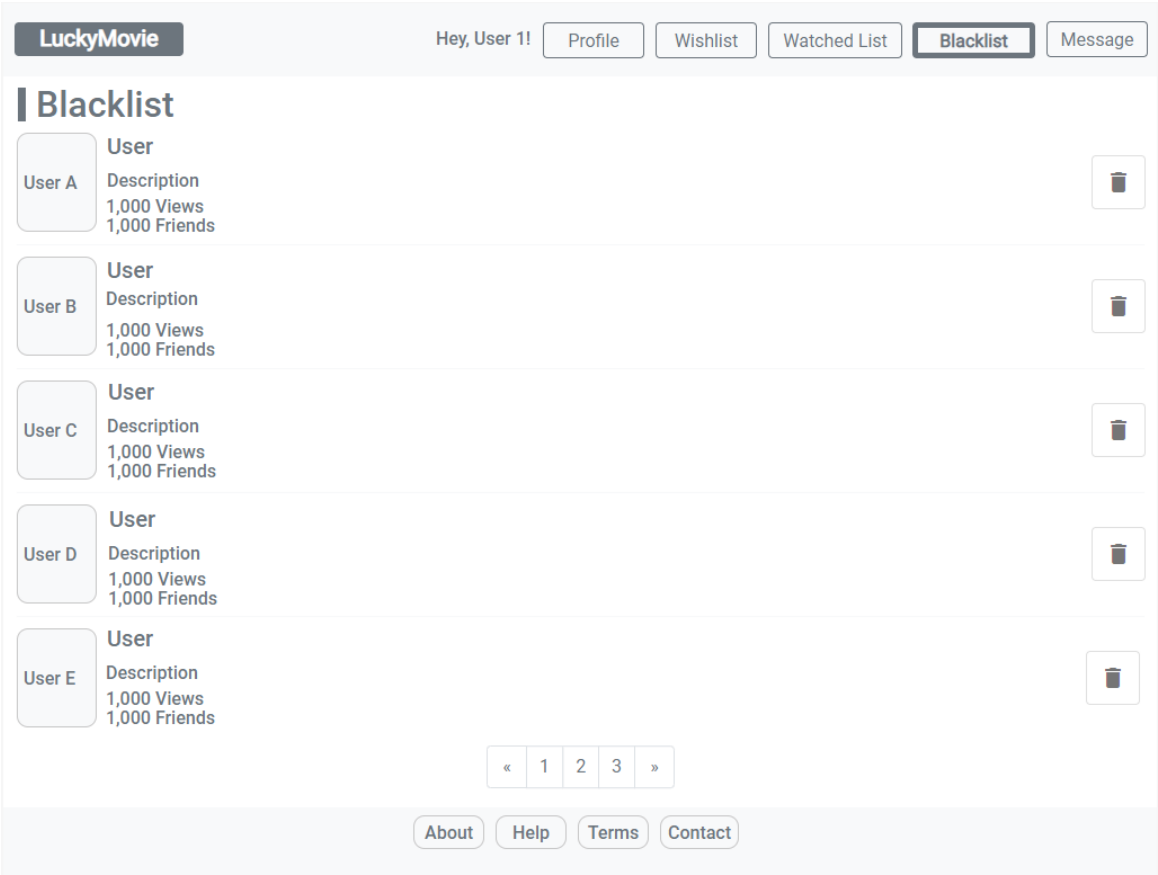
If you log in to this website, you can access your favorite movies more quickly by clicking the movie link in the wish list.

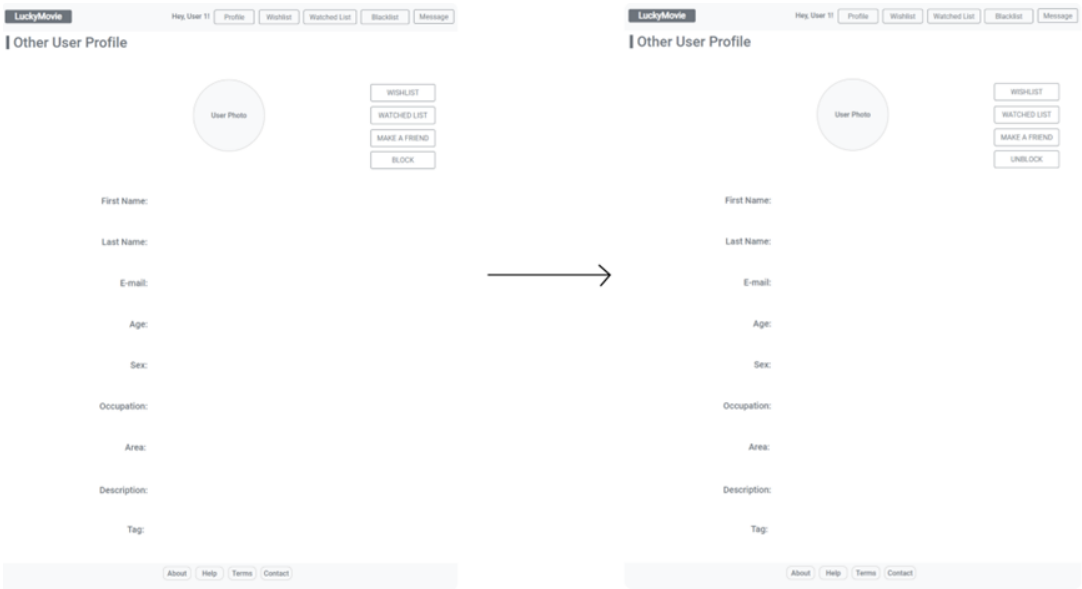


You can also set watched list.

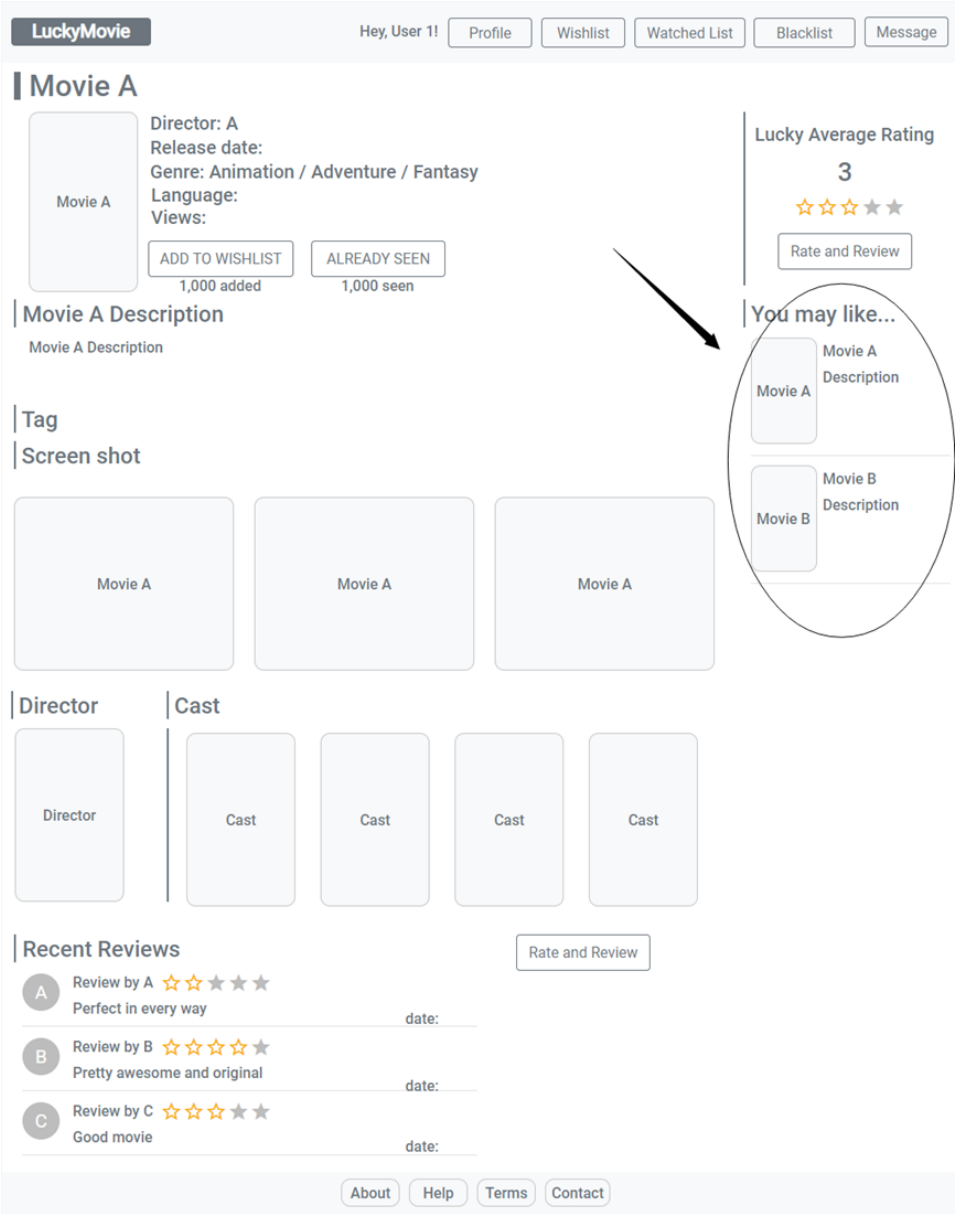


Contrary to the wish list and watched list, there is also a blacklist. As a user, if you really hate a review, you can block the person who reviewed. You can click on the current user, and there is a block button on the right side of his homepage. After you click, you will no longer see all the comments of this user. After clicking the block button, the word of the button will change to unblock. This means that if you want to unblock this user, you can click the unblock button to make this user's reviews visible to you again.

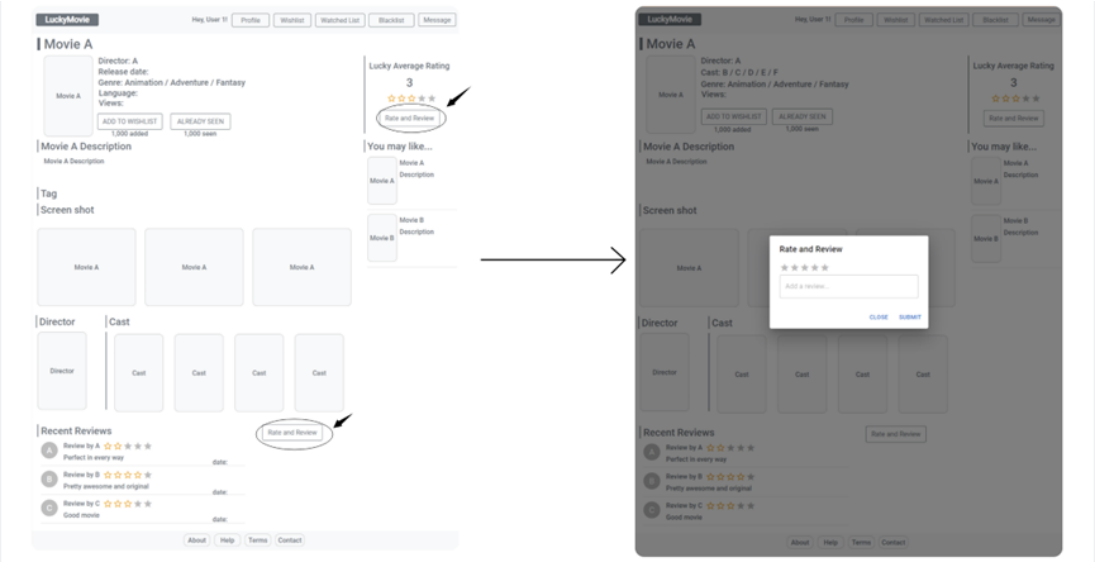




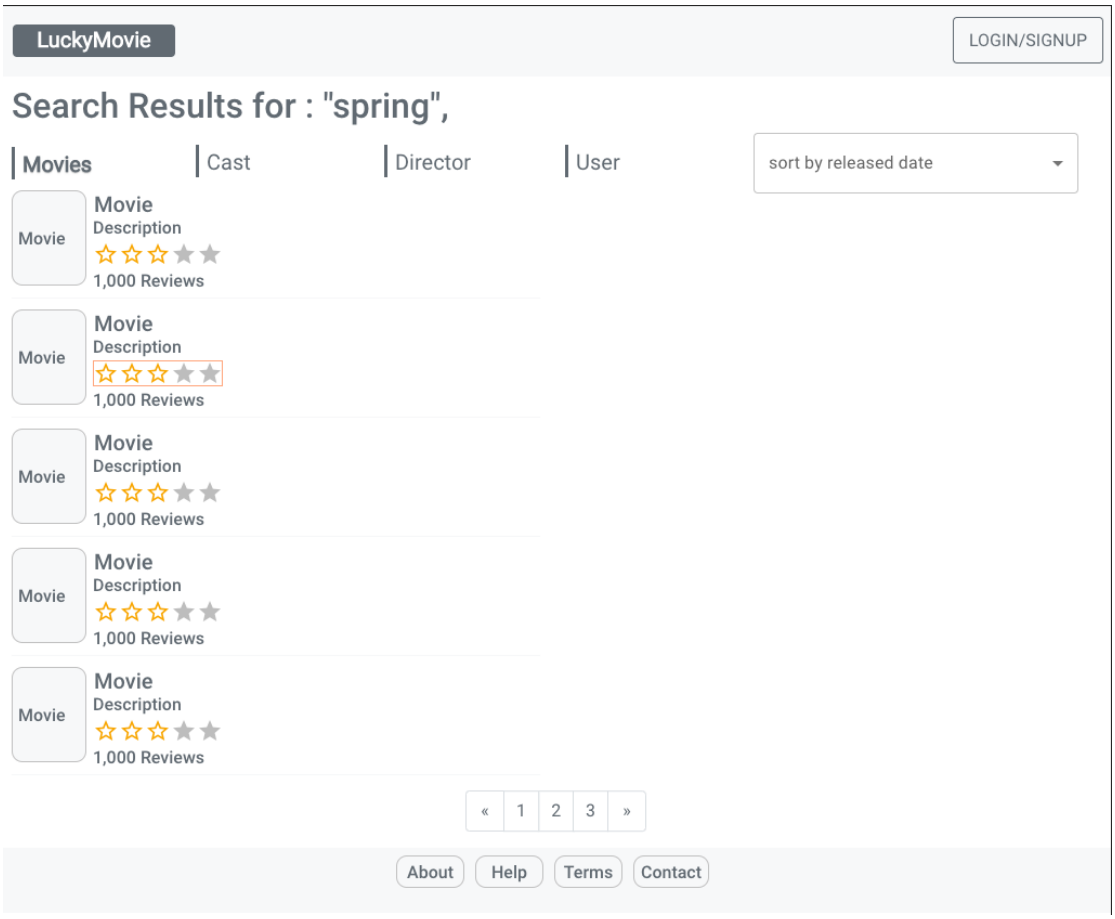
As a user of this website, the website will recommend movies for you. Users will browse and review movies, and the system will recommend them according to the tag and director name of the movies you browse and review. The recommended column will be placed on the right side of the home page.

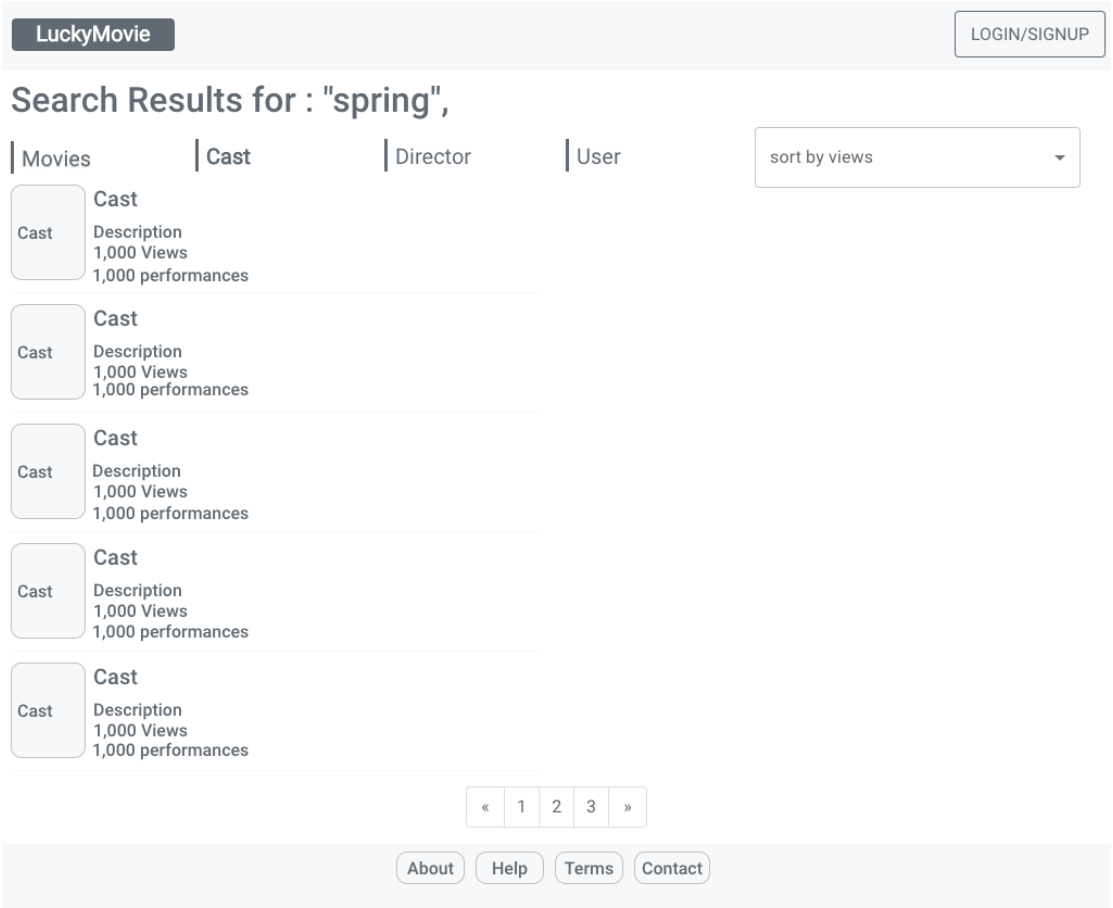


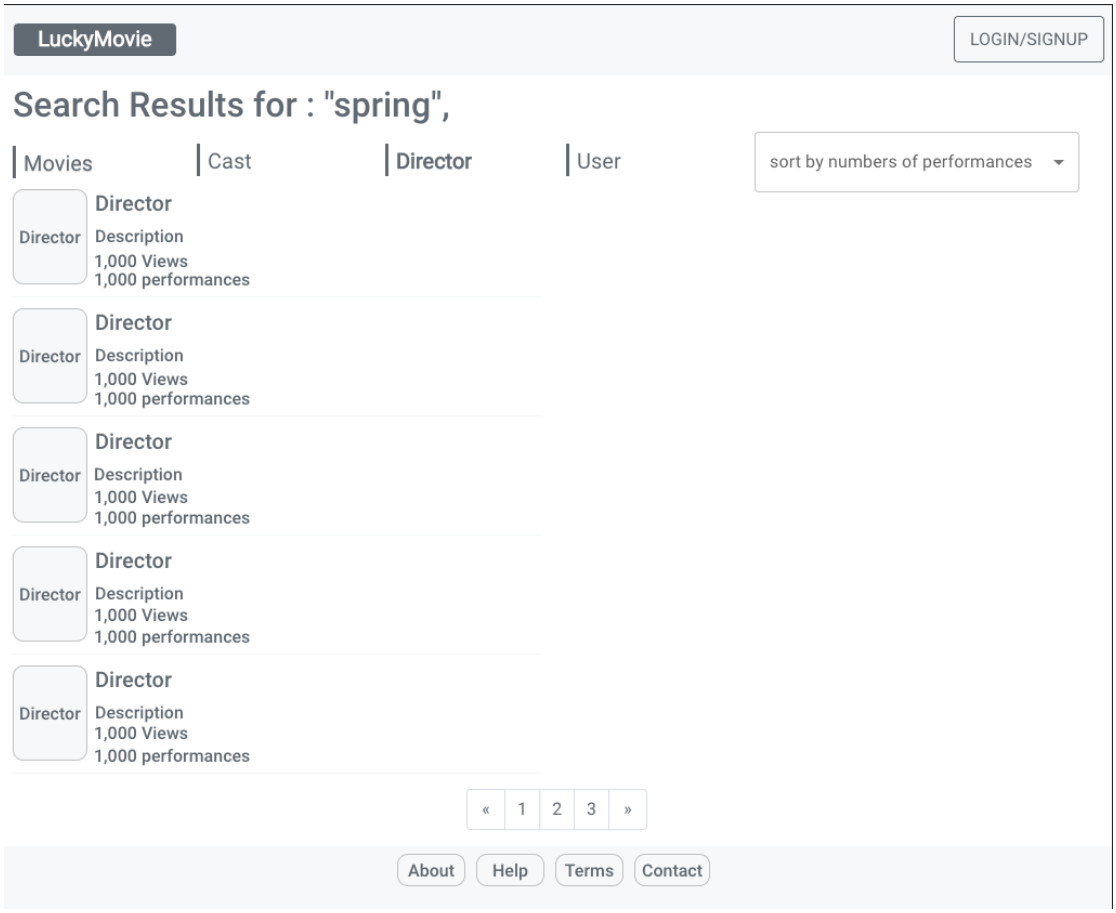
As a difference between users and tourists, users can also rate movies. The system will count how many users have rated movies and integrate these ratings. After the integration, the final rating of this movie will be made. And this rating will be placed on the introduction page of the movie.



You can search by movie, cast, director and user.




LuckyMovie
LOGIN/SIGNUP
Search Results for : "spring",
Movies
Cast
Director
User
sort by views
Cast
Cast
Description
1,000 Views
1,000 performances
« 1 2 3 »
About
Help
Terms
Contact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1,000 performances
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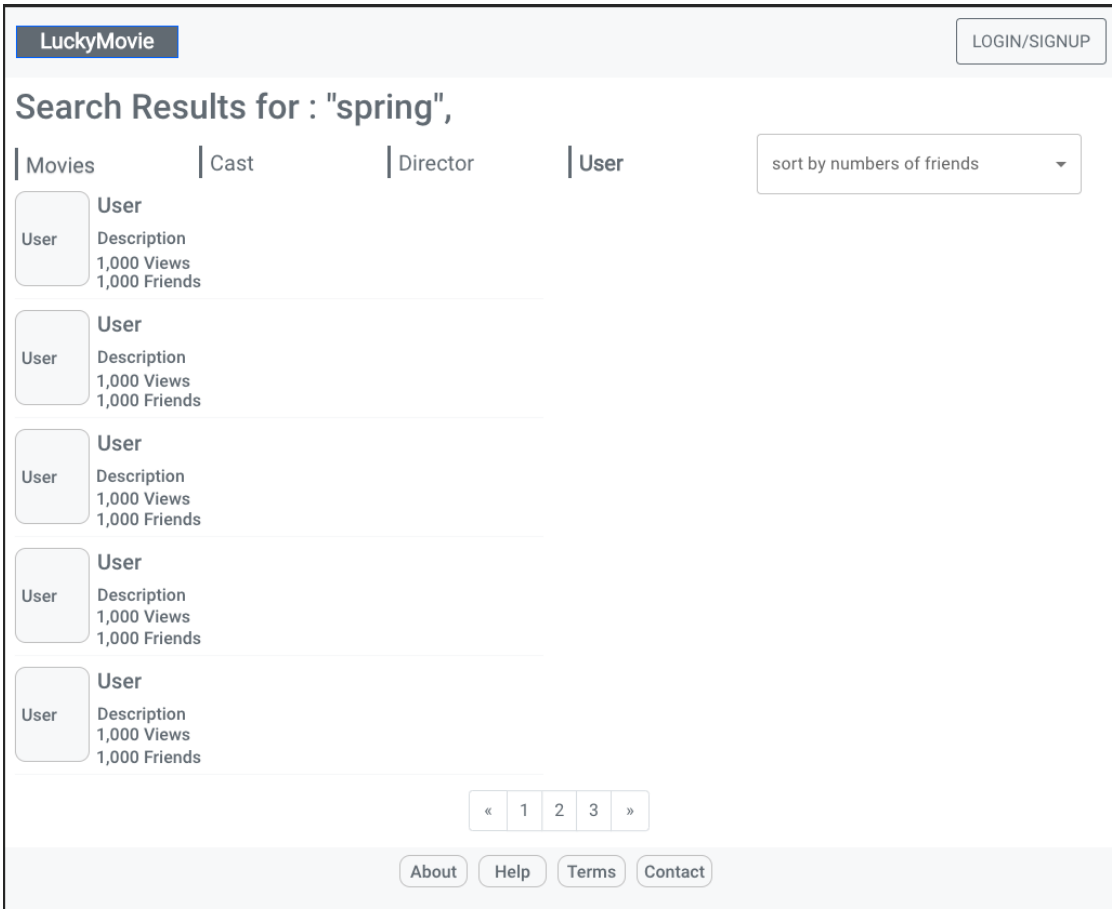
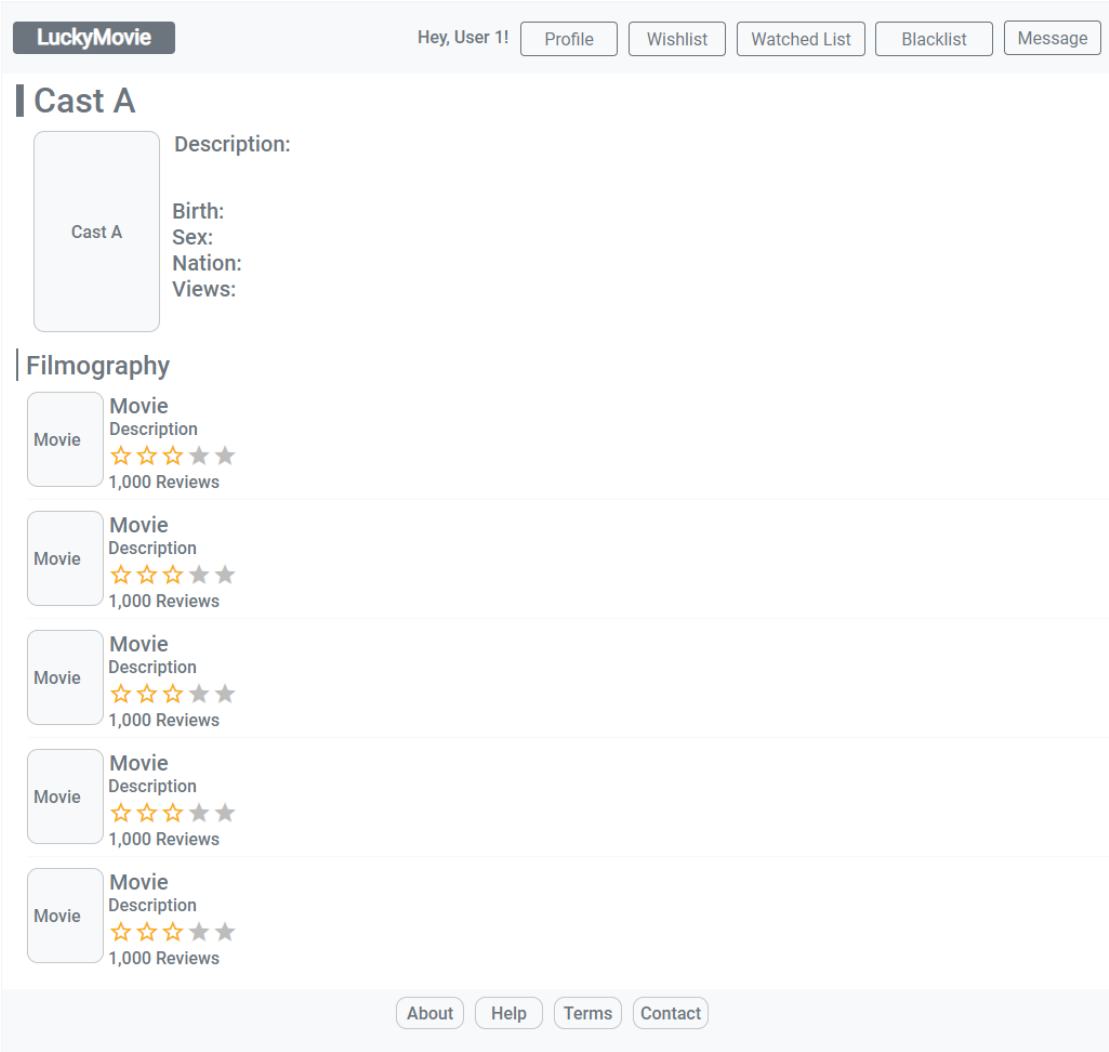
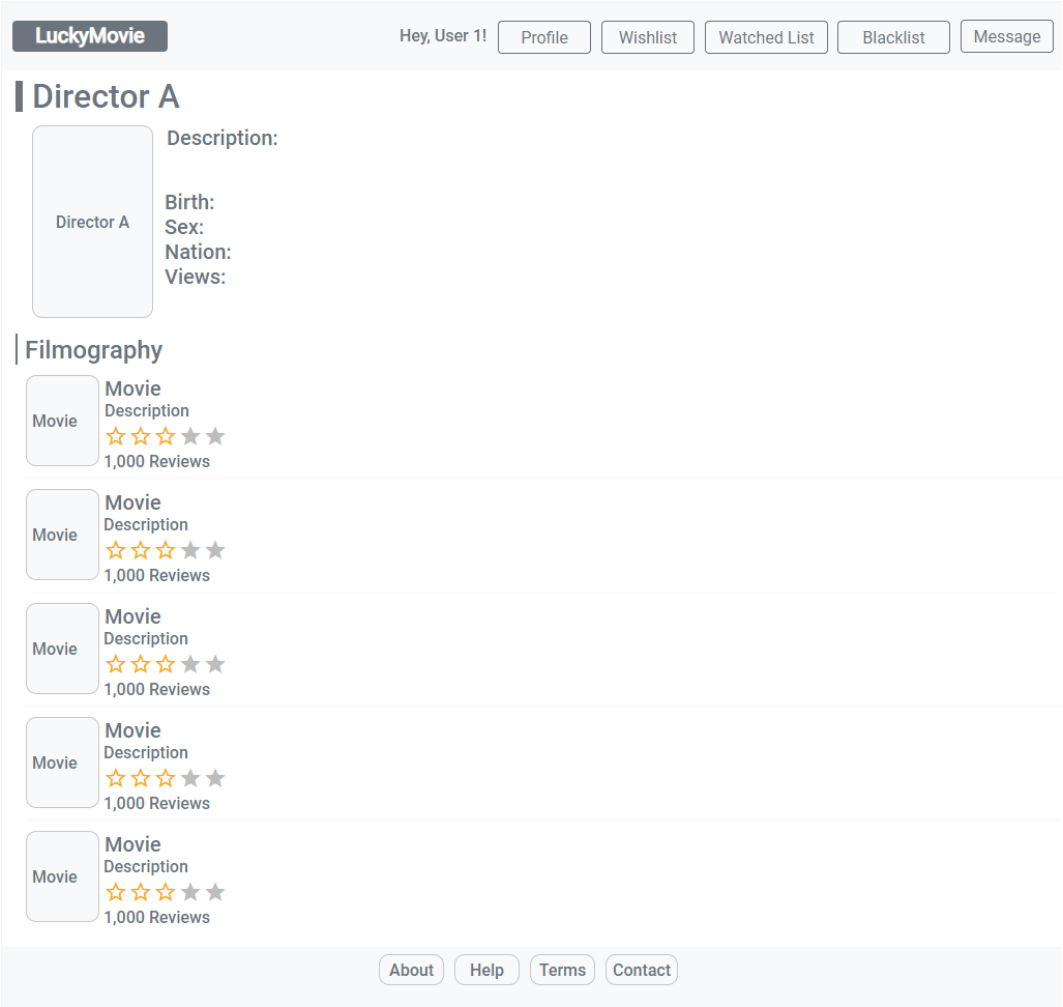


You can view the specific cast details.



You can view the specific director details.


LuckyMovie
Hey, User 1!
Profile
Wishlist
Watched List
Blacklist
Message
Director A
Director A
Description:
Birth:
Sex:
Nation:
Views:
Filmography
Movie
Movie
Description
1,000 Reviews
Movie
Movie
Description
1,000 Reviews
Movie
Movie
Description
1,000 Reviews
Movie
Movie
Description
1,000 Reviews
Movie
Movie
Description
1,000 Reviews
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